

## 1. General Description

---

The EM74LVC2GU04 is a dual unbuffered inverter. Inputs can be driven from either 3.3 V or 5 V devices. This feature allows the use of these devices as translators in mixed 3.3 V and 5 V environments.

## 2. Features and Benefits

---

- Wide supply voltage range from 1.65 V to 5.5 V
- Overvoltage tolerant inputs to 5.5 V
- High noise immunity
- $\pm 24$  mA output drive ( $V_{CC} = 3.0$  V)
- CMOS low power dissipation
- Latch-up performance exceeds 100 mA
- ESD protection:
  - HBM ANSI/ESDA/JEDEC JS-001 Class 3B exceeds 8000 V
  - MM JESD22-A115C Class C exceeds 550 V
  - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

## EM74LVC2GU04

Dual unbuffered inverter

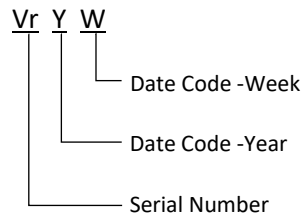
### 3. Ordering Information

Table 1. Ordering information

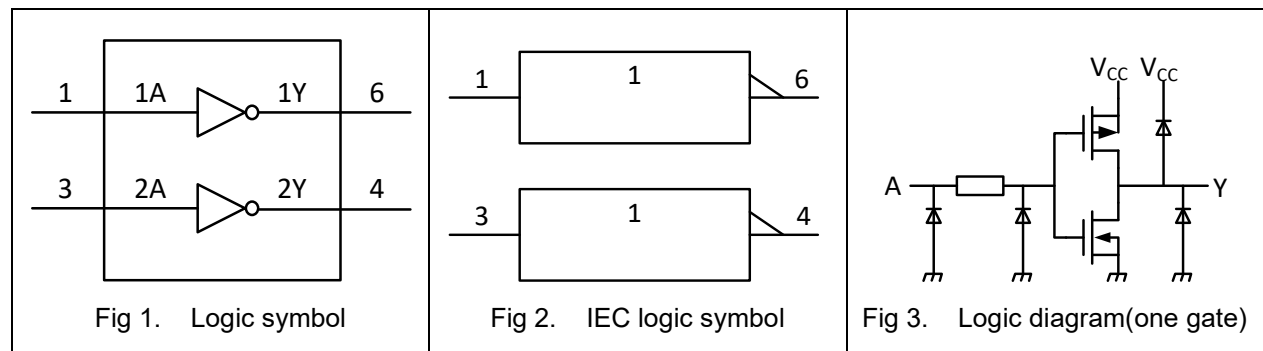
| Type number    | Topside marking | Package      |                                                                  | Quantity |
|----------------|-----------------|--------------|------------------------------------------------------------------|----------|
|                |                 | Name         | Description                                                      |          |
| EM74LVC2GU04GV | VrYW            | SOT23-6L     | SOT23 package, 6 pins<br>2.92 mm × 1.6 mm; 1.25 mm (Max) height  | 3000     |
| EM74LVC2GU04GW | VrYW            | SOT363       | SOT363 package, 6 pins<br>2.1 mm × 1.25 mm; 1.1 mm (Max) height  | 3000     |
| EM74LVC2GU04GS | Vr              | DFN1x1-6L    | DFN1×1 package, 6 pins<br>1 mm × 1 mm; 0.42 mm (Max) height      | 3000     |
| EM74LVC2GU04GM | VrYW            | DFN1x1.45-6L | DFN1.45×1 package, 6 pins<br>1.45 mm × 1 mm; 0.6 mm (Max) height | 3000     |

#### MARKING INFORMATION

NOTE: YW = Date Code.

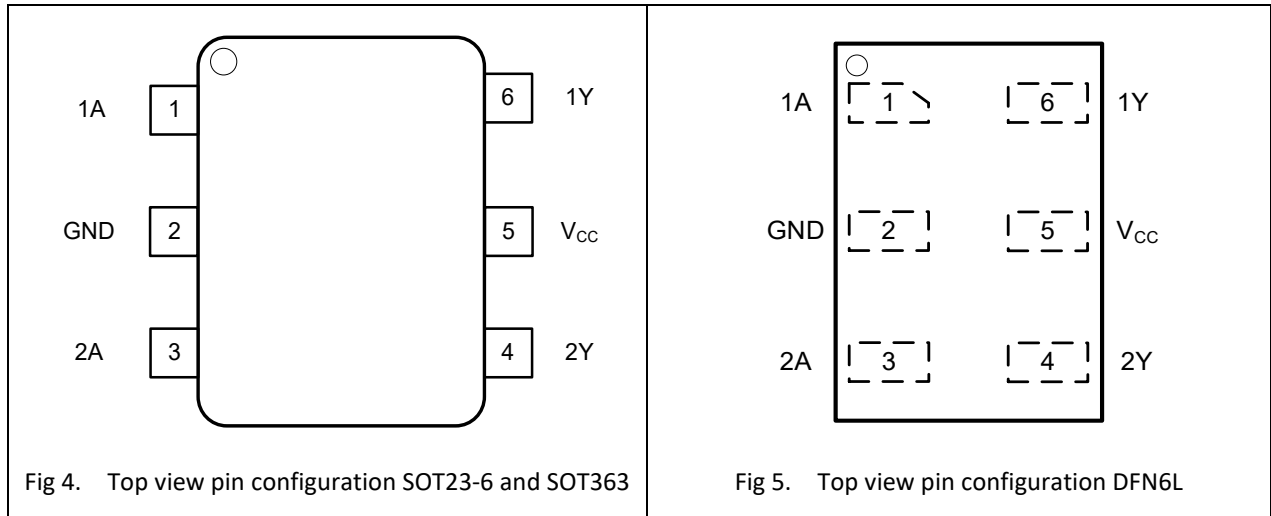


### 4. Function Diagram



## 5. Pinning Information

### 5.1. Pin map



### 5.2. Pin description

**Table 2. Pin description**

| Symbol          | Pin | Description    |
|-----------------|-----|----------------|
| 1A              | 1   | Data input     |
| GND             | 2   | Ground (0V)    |
| 2A              | 3   | Data input     |
| 2Y              | 4   | Data output    |
| V <sub>CC</sub> | 5   | Supply voltage |
| 1Y              | 6   | Data output    |

## 6. Functional Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level.

| Input | Output |
|-------|--------|
| nA    | nY     |
| L     | H      |
| H     | L      |

## 7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol    | Parameter               | Conditions                      | Min  | Max            | Unit |
|-----------|-------------------------|---------------------------------|------|----------------|------|
| $V_{CC}$  | supply voltage          |                                 | -0.5 | 6.5            | V    |
| $I_{IK}$  | input clamping current  | $V_I < 0$ V                     | -50  |                | mA   |
| $V_I$     | input voltage           | [1]                             | -0.5 | 6.5            | V    |
| $I_{OK}$  | output clamping current | $V_O > V_{CC}$ or $V_O < 0$ V   |      | $\pm 50$       | mA   |
| $V_O$     | output voltage          | Active mode [1]                 | -0.5 | $V_{CC} + 0.5$ | V    |
| $I_O$     | output current          | $V_O = 0$ V to $V_{CC}$         |      | $\pm 50$       | mA   |
| $I_{CC}$  | supply current          |                                 |      | 100            | mA   |
| $I_{GND}$ | ground current          |                                 | -100 |                | mA   |
| $P_{tot}$ | total power dissipation | $T_{amb} = -40$ °C to $+125$ °C |      | 250            | mW   |
| $T_{stg}$ | storage temperature     |                                 | -65  | 150            | °C   |

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

## 8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol              | Parameter                           | Conditions                        | Min  | Typ | Max             | Unit |
|---------------------|-------------------------------------|-----------------------------------|------|-----|-----------------|------|
| V <sub>CC</sub>     | supply voltage                      |                                   | 1.65 |     | 5.5             | V    |
| V <sub>I</sub>      | input voltage                       |                                   | 0    |     | 5.5             | V    |
| V <sub>O</sub>      | output voltage                      | Active mode                       | 0    |     | V <sub>CC</sub> | V    |
| T <sub>amb</sub>    | ambient temperature                 |                                   | -40  |     | 125             | °C   |
| $\Delta t/\Delta V$ | Input transition rise and fall rate | V <sub>CC</sub> = 1.65 V to 2.7 V |      |     | 20              | ns/V |
|                     |                                     | V <sub>CC</sub> = 2.7 V to 5.5 V  |      |     | 10              | ns/V |

## 9.Static Characteristics

**Table 6. Static characteristics**

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

| Symbol   | Parameter                 | Conditions                                                           | -40 °C to +85 °C |           |              | -40 °C to +125 °C |              | Unit          |
|----------|---------------------------|----------------------------------------------------------------------|------------------|-----------|--------------|-------------------|--------------|---------------|
|          |                           |                                                                      | Min              | Typ[1]    | Max          | Min               | Max          |               |
| $V_{IH}$ | HIGH-level input voltage  | $V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$                         | $0.75V_{CC}$     |           |              | $0.8V_{CC}$       |              | V             |
|          |                           | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$                           | $0.75V_{CC}$     |           |              | $0.8V_{CC}$       |              | V             |
|          |                           | $V_{CC} = 2.7 \text{ V to } 3.6 \text{ V}$                           | $0.75V_{CC}$     |           |              | $0.8V_{CC}$       |              | V             |
|          |                           | $V_{CC} = 4.5 \text{ V to } 5.5 \text{ V}$                           | $0.8V_{CC}$      |           |              | $0.8V_{CC}$       |              | V             |
| $V_{IL}$ | LOW-level input voltage   | $V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$                         |                  |           | $0.25V_{CC}$ |                   | $0.2V_{CC}$  | V             |
|          |                           | $V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$                           |                  |           | $0.25V_{CC}$ |                   | $0.2V_{CC}$  | V             |
|          |                           | $V_{CC} = 2.7 \text{ V to } 3.6 \text{ V}$                           |                  |           | $0.24V_{CC}$ |                   | $0.2V_{CC}$  | V             |
|          |                           | $V_{CC} = 4.5 \text{ V to } 5.5 \text{ V}$                           |                  |           | $0.16V_{CC}$ |                   | $0.15V_{CC}$ | V             |
| $V_{OH}$ | HIGH-level output voltage | $V_I = V_{IH} \text{ or } V_{IL}$                                    |                  |           |              |                   |              |               |
|          |                           | $I_O = -100\mu\text{A}; V_{CC} = 1.65 \text{ V to } 5.5 \text{ V}$   | $V_{CC} - 0.1$   |           |              | $V_{CC} - 0.1$    |              | V             |
|          |                           | $I_O = -4 \text{ mA}; V_{CC} = 1.65 \text{ V}$                       | 1.2              |           |              | 0.95              |              | V             |
|          |                           | $I_O = -8 \text{ mA}; V_{CC} = 2.3 \text{ V}$                        | 1.9              |           |              | 1.7               |              | V             |
|          |                           | $I_O = -12 \text{ mA}; V_{CC} = 2.7 \text{ V}$                       | 2.2              |           |              | 1.9               |              | V             |
|          |                           | $I_O = -24 \text{ mA}; V_{CC} = 3.0 \text{ V}$                       | 2.3              |           |              | 2.0               |              | V             |
|          |                           | $I_O = -32 \text{ mA}; V_{CC} = 4.5 \text{ V}$                       | 3.8              |           |              | 3.4               |              | V             |
| $V_{OL}$ | LOW-level output voltage  | $V_I = V_{IH} \text{ or } V_{IL}$                                    |                  |           |              |                   |              |               |
|          |                           | $I_O = 100\mu\text{A}; V_{CC} = 1.65 \text{ V to } 5.5 \text{ V}$    |                  |           | 0.10         |                   | 0.10         | V             |
|          |                           | $I_O = 4 \text{ mA}; V_{CC} = 1.65 \text{ V}$                        |                  |           | 0.45         |                   | 0.70         | V             |
|          |                           | $I_O = 8 \text{ mA}; V_{CC} = 2.3 \text{ V}$                         |                  |           | 0.30         |                   | 0.45         | V             |
|          |                           | $I_O = 12 \text{ mA}; V_{CC} = 2.7 \text{ V}$                        |                  |           | 0.40         |                   | 0.60         | V             |
|          |                           | $I_O = 24 \text{ mA}; V_{CC} = 3.0 \text{ V}$                        |                  |           | 0.55         |                   | 0.80         | V             |
|          |                           | $I_O = 32 \text{ mA}; V_{CC} = 4.5 \text{ V}$                        |                  |           | 0.55         |                   | 0.80         | V             |
| $I_i$    | Input leakage current     | $V_I = 5.5 \text{ V or GND}; V_{CC} = 0 \text{ V to } 5.5 \text{ V}$ |                  | $\pm 0.1$ | $\pm 1$      |                   | $\pm 1$      | $\mu\text{A}$ |

## EM74LVC2GU04

### Dual unbuffered inverter

| Symbol          | Parameter         | Conditions                                                                              | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|-----------------|-------------------|-----------------------------------------------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|                 |                   |                                                                                         | Min              | Typ[1] | Max | Min               | Max |      |
| I <sub>CC</sub> | supply current    | V <sub>I</sub> = 5.5V or GND ; I <sub>O</sub> = 0A ;<br>V <sub>CC</sub> = 1.65V to 5.5V |                  | 0.1    | 4   |                   | 4   | μA   |
| C <sub>I</sub>  | input capacitance | V <sub>CC</sub> = 3.3V ; V <sub>I</sub> = GND to V <sub>CC</sub>                        |                  | 5.3    |     |                   |     | pF   |

[1] All typical values are measured at V<sub>CC</sub> = 3.3V and T<sub>amb</sub> = 25°C.

## 10. Dynamic Characteristics

**Table 7. Dynamic characteristics**

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 7.

| Symbol          | Parameter                     | Conditions                                                           | -40 °C to +85 °C |        |     | -40 °C to +125 °C |     | Unit |
|-----------------|-------------------------------|----------------------------------------------------------------------|------------------|--------|-----|-------------------|-----|------|
|                 |                               |                                                                      | Min              | Typ[1] | Max | Min               | Max |      |
| t <sub>pd</sub> | propagation delay             | nA to nY; see Fig. 6 [2]                                             |                  |        |     |                   |     |      |
|                 |                               | V <sub>CC</sub> = 1.65 V to 1.95 V                                   | 0.8              | 1.6    | 3.3 | 0.8               | 3.4 | ns   |
|                 |                               | V <sub>CC</sub> = 2.3 V to 2.7 V                                     | 0.7              | 1.4    | 2.6 | 0.7               | 2.7 | ns   |
|                 |                               | V <sub>CC</sub> = 3.0 V to 3.6 V                                     | 0.8              | 1.3    | 2.2 | 0.8               | 2.4 | ns   |
|                 |                               | V <sub>CC</sub> = 4.5 V to 5.5 V                                     | 0.7              | 1.3    | 2.0 | 0.7               | 2.1 | ns   |
| C <sub>PD</sub> | power dissipation capacitance | V <sub>I</sub> = GND to V <sub>CC</sub> ; V <sub>CC</sub> = 3.3V [3] |                  | 12.1   |     |                   |     | pF   |

[1] Typical values are measured at T<sub>amb</sub> = 25 °C and V<sub>CC</sub> = 1.8 V, 2.5 V, 3.3 V and 5.0 V respectively.

[2] t<sub>pd</sub> is the same as t<sub>PLH</sub> and t<sub>PHL</sub>.

[3] C<sub>PD</sub> is used to determine the dynamic power dissipation (P<sub>D</sub> in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$  where:

f<sub>i</sub> = input frequency in MHz;

f<sub>o</sub> = output frequency in MHz;

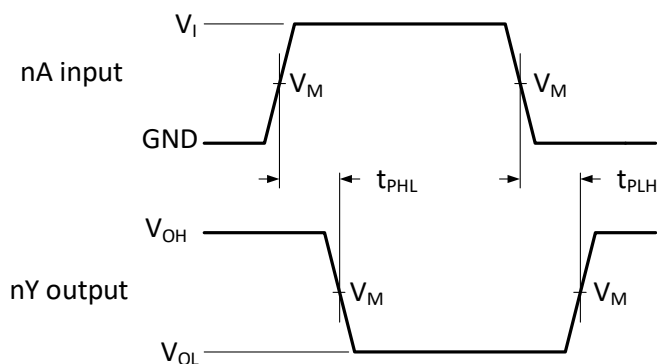
C<sub>L</sub> = output load capacitance in pF;

V<sub>CC</sub> = supply voltage in V;

N = number of inputs switching;

$\sum (C_L \times V_{CC}^2 \times f_o)$  = sum of outputs.

## 10.1. Waveforms and test circuit



Measurement points are given in Table 8.

$V_{OL}$  and  $V_{OH}$  are typical output voltage levels that occur with the output load.

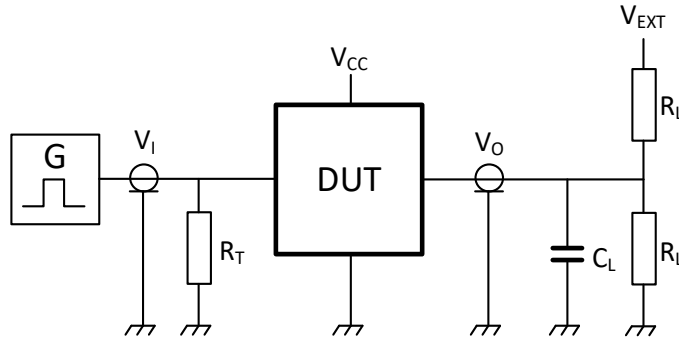
Fig 6. The input nA to output nY propagation delays

**Table 8. Measurement points**

| Supply voltage   | Input       | Output      |
|------------------|-------------|-------------|
| $V_{CC}$         | $V_M$       | $V_M$       |
| 1.65 V to 1.95 V | $0.5V_{CC}$ | $0.5V_{CC}$ |
| 2.3 V to 2.7 V   | $0.5V_{CC}$ | $0.5V_{CC}$ |
| 3.0 V to 3.6 V   | 1.5 V       | 1.5 V       |
| 4.5 V to 5.5 V   | $0.5V_{CC}$ | $0.5V_{CC}$ |

## EM74LVC2GU04

### Dual unbuffered inverter



Test data is given in Table 9.

Definitions for test circuit:

$R_L$  = Load resistance.

$C_L$  = Load capacitance including jig and probe capacitance.

$R_T$  = Termination resistance should be equal to the output impedance  $Z_o$  of the pulse generator.

$V_{EXT}$  = External voltage for measuring switching times.

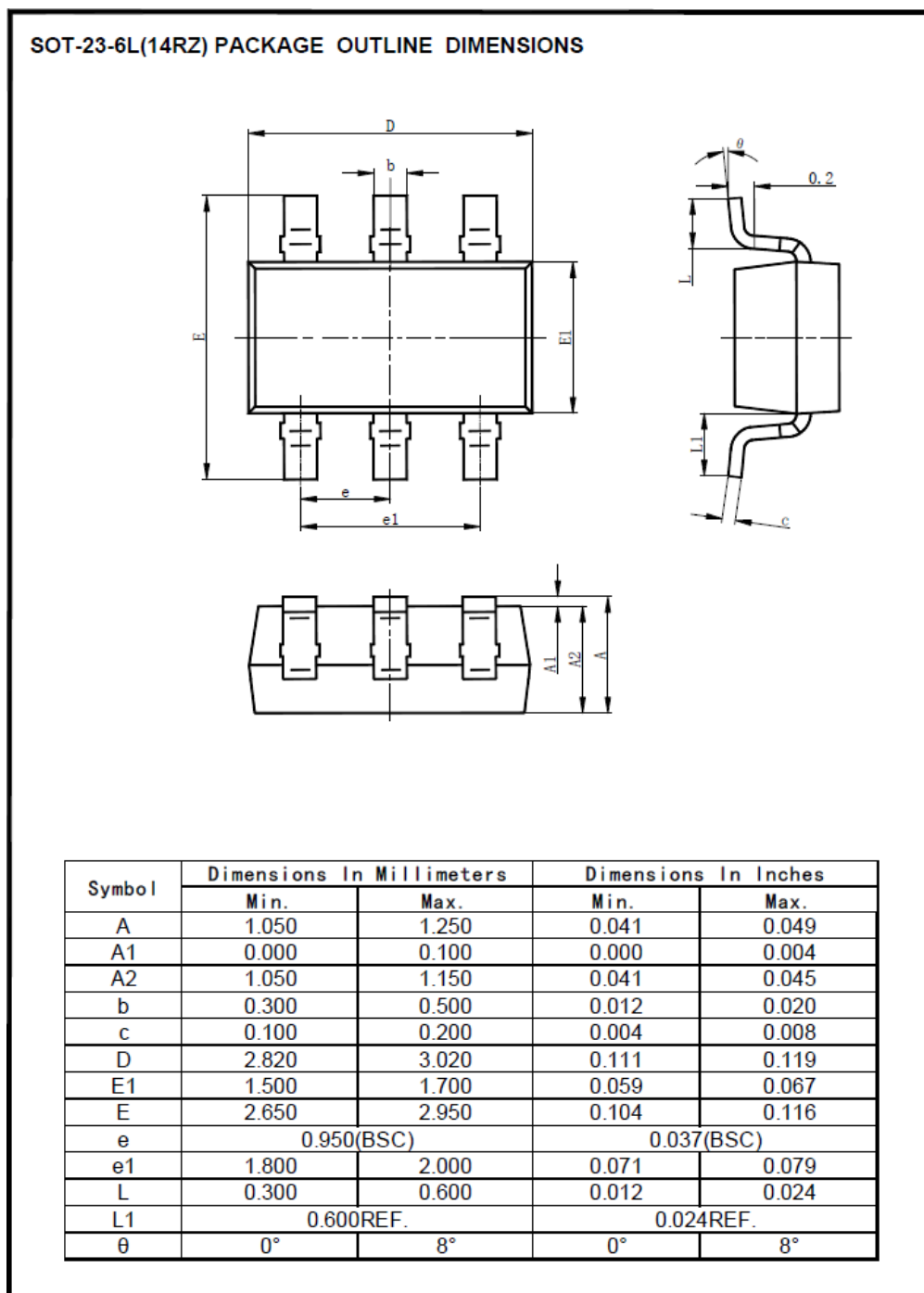
Fig 7. Test circuit for measuring switching times

Table 9. Test data

| Supply voltage   | Input    |               | Load  |              | $V_{EXT}$          |
|------------------|----------|---------------|-------|--------------|--------------------|
| $V_{CC}$         | $V_I$    | $t_r = t_f$   | $C_L$ | $R_L$        | $t_{PLH}, t_{PHL}$ |
| 1.65 V to 1.95 V | $V_{CC}$ | $\leq 2.0$ ns | 30 pF | 1 k $\Omega$ | open               |
| 2.3 V to 2.7 V   | $V_{CC}$ | $\leq 2.0$ ns | 30 pF | 500 $\Omega$ | open               |
| 3.0 V to 3.6 V   | 3 V      | $\leq 2.5$ ns | 50 pF | 500 $\Omega$ | open               |
| 4.5 V to 5.5 V   | $V_{CC}$ | $\leq 2.5$ ns | 50 pF | 500 $\Omega$ | open               |

## 11. Package Outline

### SOT23-6L

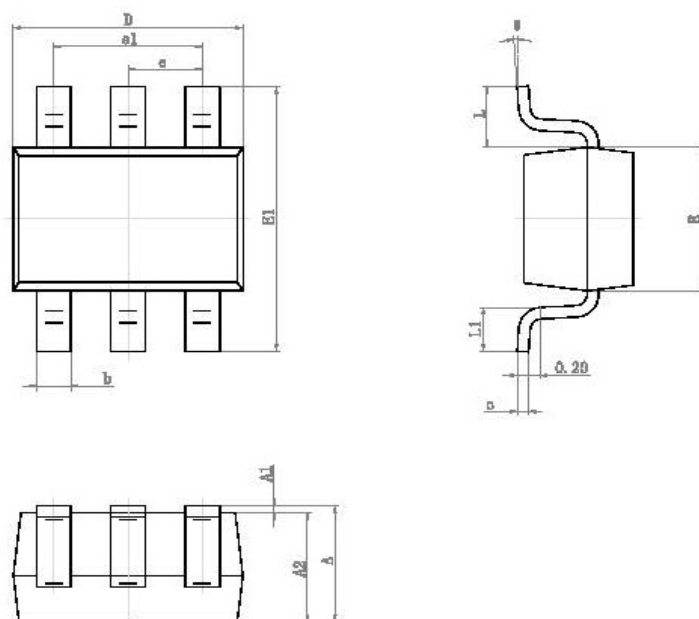


# EM74LVC2GU04

Dual unbuffered inverter

SOT363

## SOT-363 (16R) PACKAGE OUTLINE DIMENSIONS

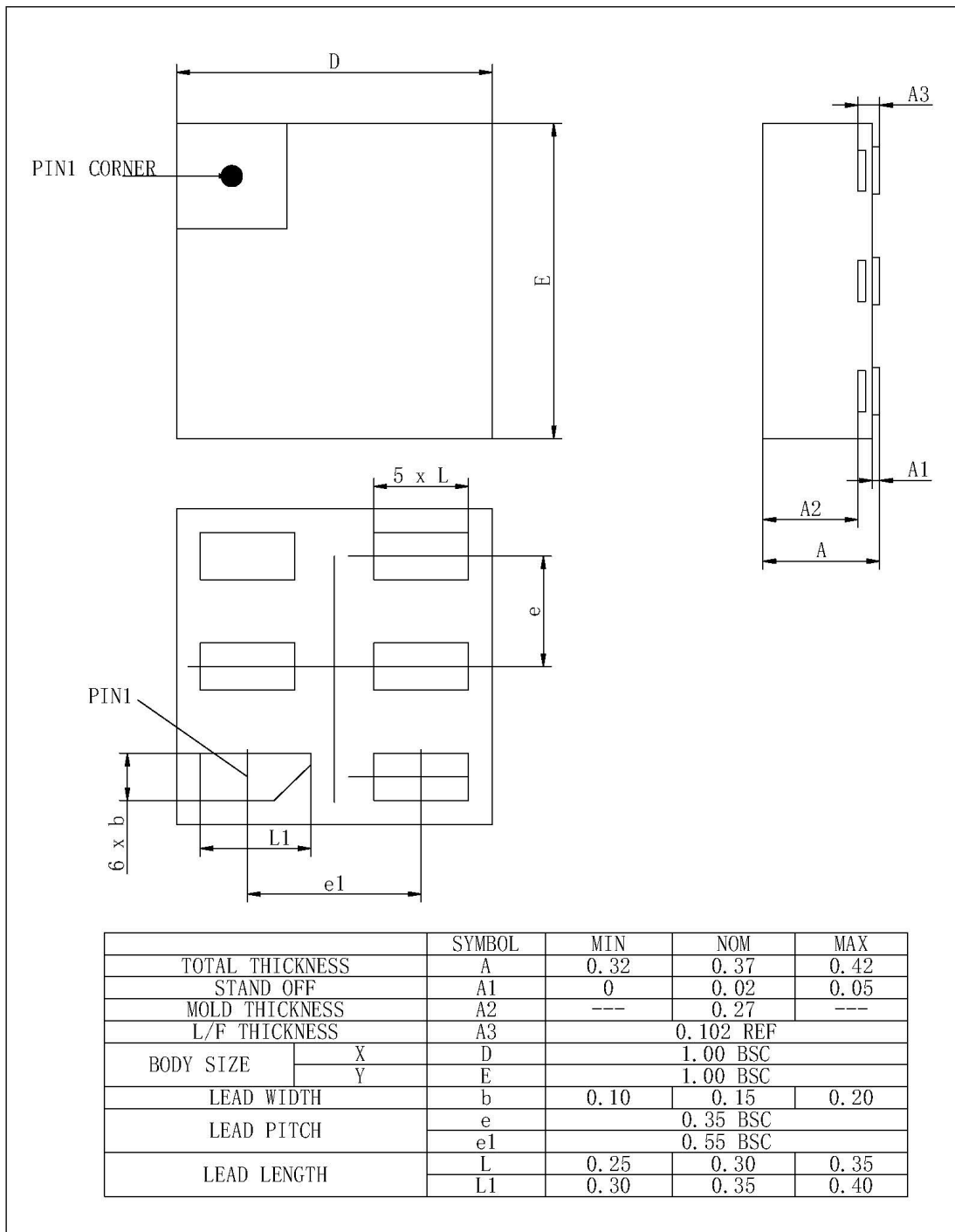


| Symbol | Dimensions In Millimeters |       | Dimensions In Inches |       |
|--------|---------------------------|-------|----------------------|-------|
|        | Min.                      | Max.  | Min.                 | Max.  |
| A      | 0.900                     | 1.100 | 0.035                | 0.043 |
| A1     | 0.000                     | 0.100 | 0.000                | 0.004 |
| A2     | 0.900                     | 1.000 | 0.035                | 0.039 |
| b      | 0.150                     | 0.350 | 0.006                | 0.014 |
| c      | 0.110                     | 0.175 | 0.004                | 0.007 |
| D      | 2.000                     | 2.200 | 0.079                | 0.087 |
| E      | 1.150                     | 1.350 | 0.045                | 0.053 |
| E1     | 2.150                     | 2.450 | 0.085                | 0.096 |
| e      | 0.650 TYP.                |       | 0.026 TYP.           |       |
| e1     | 1.200                     | 1.400 | 0.047                | 0.055 |
| L      | 0.525 REF.                |       | 0.021 REF.           |       |
| L1     | 0.260                     | 0.460 | 0.010                | 0.018 |
| θ      | 0°                        | 8°    | 0°                   | 8°    |

# EM74LVC2GU04

Dual unbuffered inverter

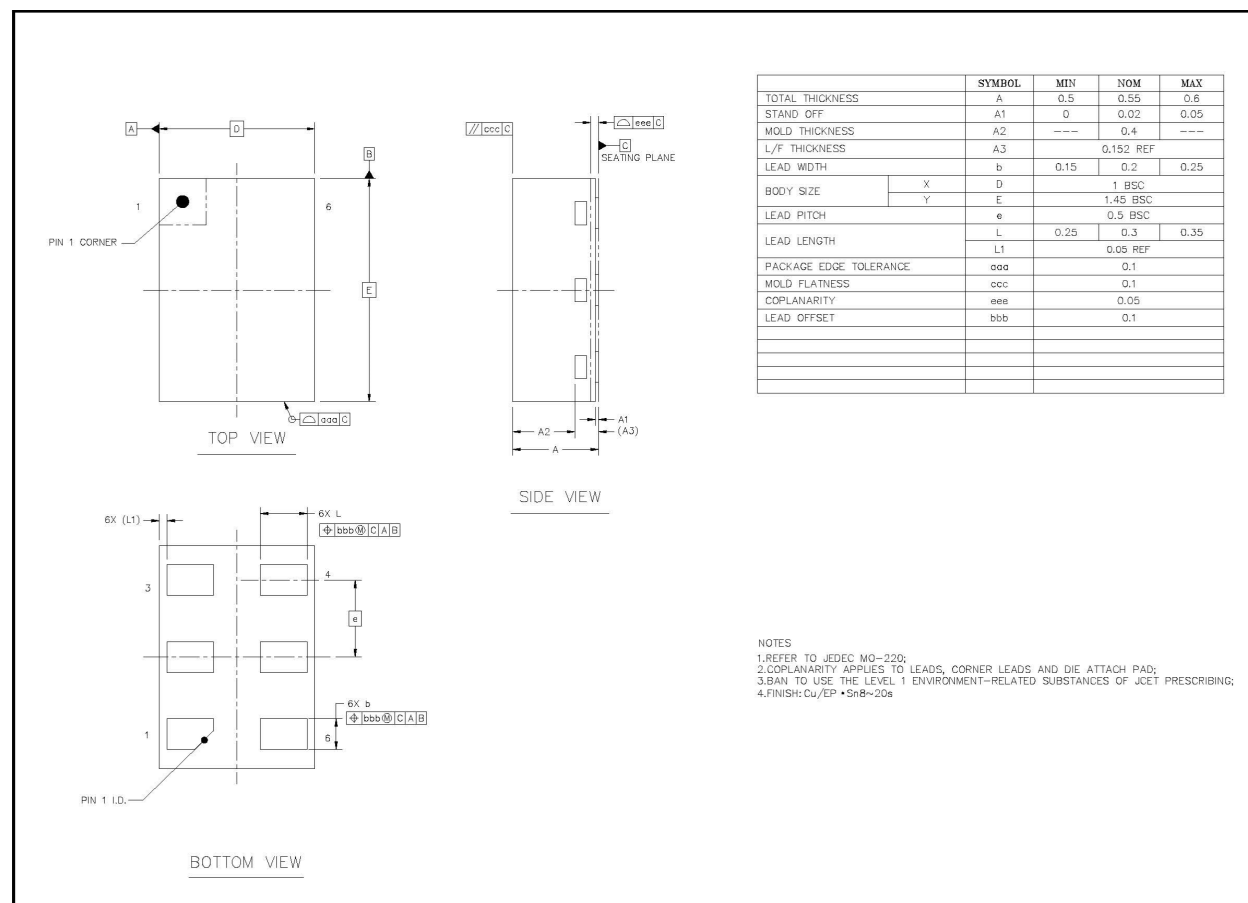
## DFN1x1-6L



# EM74LVC2GU04

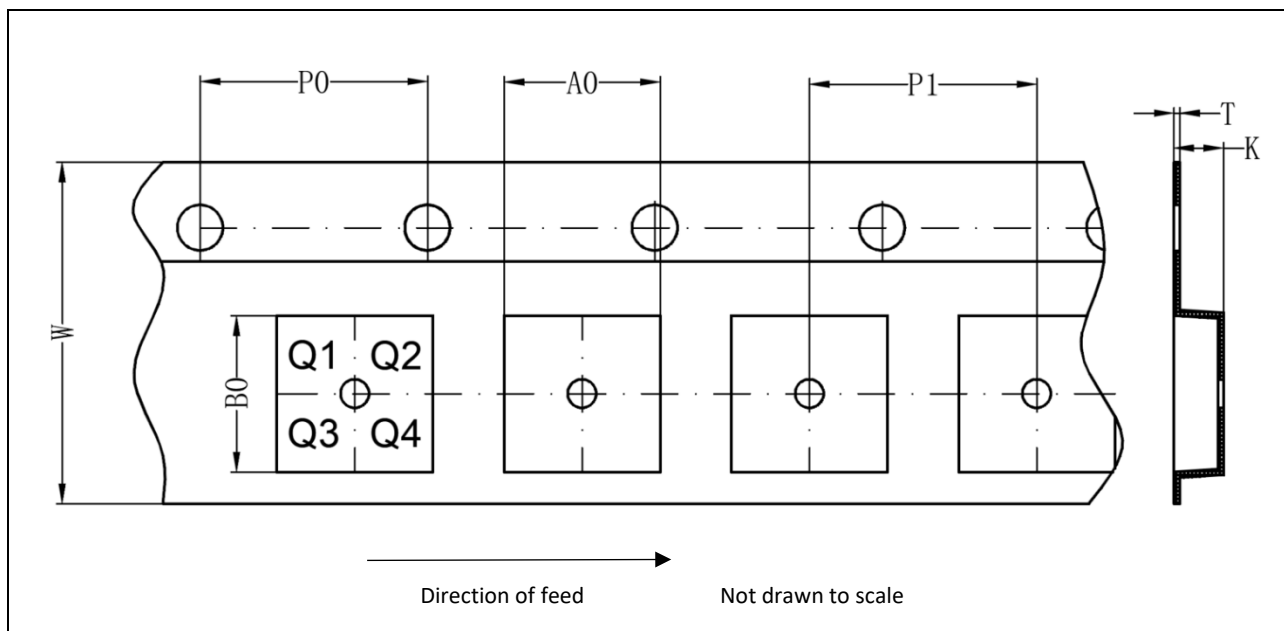
Dual unbuffered inverter

## DFN1x1.45-6L



## 12. Tape and Reel Information

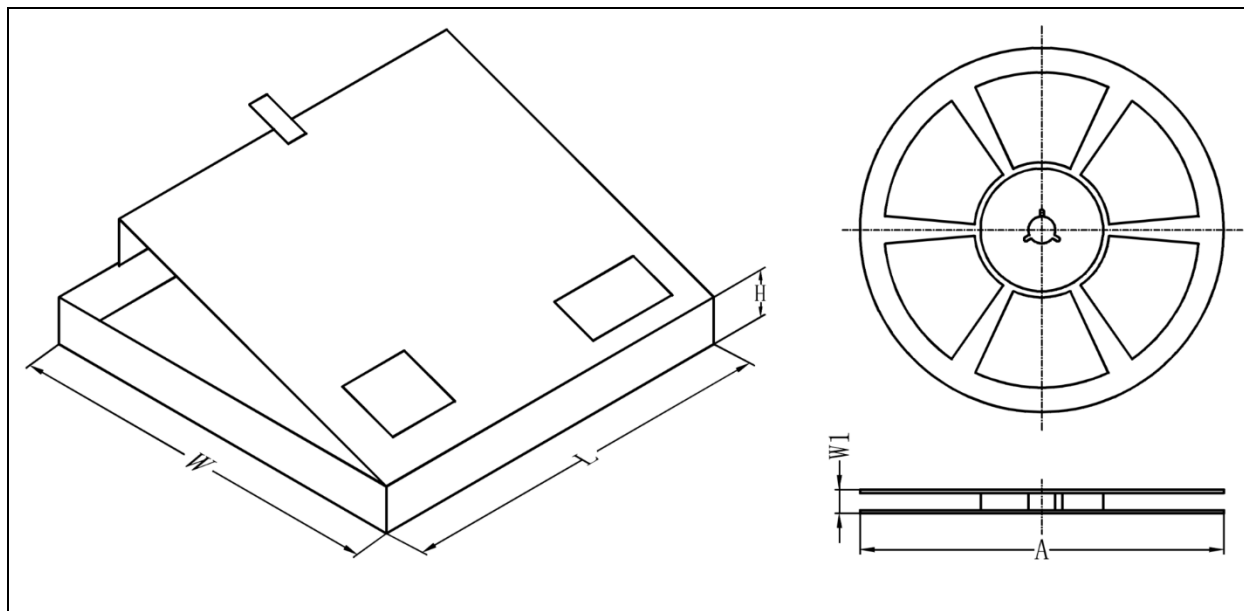
### 12.1. Carrier tape dimensions



**Table 10. Carrier tape dimensions**

| Package version | A0(mm) | B0(mm) | K(mm) | T(mm) | P1(mm) | W(mm) | P0(mm) | PIN 1 |
|-----------------|--------|--------|-------|-------|--------|-------|--------|-------|
| SOT23-6L        | 3.23   | 3.17   | 1.37  | 0.25  | 4      | 8     | 4      | Q3    |
| SOT363          | 2.55   | 2.55   | 1.2   | 0.2   | 4      | 8     | 4      | Q3    |
| DFN1x1-6L       | 1.16   | 1.16   | 0.5   | 0.23  | 4      | 8     | 4      | Q1    |
| DFN1x1.45-6L    | 1.15   | 1.6    | 0.75  | 0.2   | 4      | 8     | 4      | Q1    |

## 12.2. Reel and box dimensions



**Table11.Dimensions and quantities**

| Package version | Type NO. ending | Reel Dimension A (mm) | Reel Width W1 (mm) | SPQ (pcs)[1] | Reels per box | Outer box dimensions L×W×H(mm)[2] |
|-----------------|-----------------|-----------------------|--------------------|--------------|---------------|-----------------------------------|
| SOT23-6L        | GV              | 180                   | 12.2               | 3000         | 1             | 210x200x40                        |
| SOT363          | GW              | 180                   | 12.2               | 3000         | 1             | 210x200x40                        |
| DFN1x1-6L       | GS              | 180                   | 12.2               | 3000         | 1             | 210x200x40                        |
| DFN1x1.45-6L    | GM              | 180                   | 12.2               | 3000         | 1             | 210x200x40                        |

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

## 13. Abbreviations

**Table 12. Abbreviations**

| Acronym | Description                             |
|---------|-----------------------------------------|
| CMOS    | Complementary Metal-Oxide Semiconductor |
| DUT     | Device Under Test                       |
| ESD     | Electro Static Discharge                |
| HBM     | Human Body Model                        |
| MM      | Machine Model                           |
| TTL     | Transistor-Transistor Logic             |

## 14. Revision History

**Table 13. Revision history**

| Document ID         | Release Date                                                                                                                                         | Data sheet status | Change notice | Supersedes          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------|---------------------|
| EM74LVC2GU04 Rev1.1 | Dec 15, 2025                                                                                                                                         | Product datasheet |               | EM74LVC2GU04 Rev1.0 |
| Modifications:      | <ul style="list-style-type: none"><li>• Table 1: Added topside marking information.</li><li>• Section 12: Added tape and reel information.</li></ul> |                   |               |                     |
| EM74LVC2GU04 Rev1.0 | Oct 30, 2023                                                                                                                                         | Product datasheet |               |                     |